MATH0119 Dynamical Systems

Year:	2026–2027
Code:	MATH0119
Level:	7 (UG/PG)
Normal student group(s):	UG: Year 4 or Year 3 Mathematics degrees PG: MSc Mathematics
Value:	15 credits (= 7.5 ECTS credits)
Term:	2
Assessment:	90% examination, 10% coursework
Normal Pre-requisites:	MATH0017 and (MATH0019 or MATH0020)
Lecturer:	Dr S Ghazouani

Course Description and Objectives

In this course, we will give an introduction to the theory of dynamical systems. A dynamical system is given by a transformation of a topological space which we think of as the law driving the evolution of a system over time (such as the evolution over time of the solar system, a population, the weather, you-name-it). Our goal will be to develop the tools and the language which allow for the general treatment of such problems: this will involve a good dose of topology, analysis and measure-theory.

Recommended Texts

1. Introduction to the Modern Theory of Dynamical Systems, by Katok and Hasselblatt
2. Introduction to Dynamical Systems, by Brin and Stuck
3. Ergodic theory: with a view towards number theory, by Einsiedler and Ward

Detailed Syllabus

– **Quasi-periodic dynamics.** Linear flows on the torus, circle homeomorphisms, rotation number, Poincaré theorem.

– **Chaotic dynamics.** Doubling map, shift maps, Smale's horseshoe

– **Hyperbolic dynamics.** Hyperbolic fixed points in $\mathbb{R}^2$, Stable Curve Theorem, Local Stability,homoclinic intersections

– **Ergodic theory.** Invariant measures, Ergodicity, Birkhoff's Ergodic Theorem